

Chapter 123 — Sphincteric Incontinence in the Male: AUS and Slings — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **Artificial urinary sphincter (AUS) = gold standard**, every stress urinary incontinence (SUI) severity; works after radiotherapy (RT) and without detrusor contraction.
2. **24-h pad weight: mild <150 g, moderate <400 g** → slings for mild-moderate.
3. **Severe UI or any RT → AUS first-line** (AUA/SUFU guideline).
4. **MASTER trial**: sling non-inferior overall; **AUS better if >250 g/24 h or prior RT**.
5. **Bulbar cuff proximal to corporal fusion**; usually 4.0–4.5 cm.
6. **Pressure-regulating balloon (PRB) 61–70 cm H₂O bulbar**; 51–60 fragile; 71–80 bladder neck; fill 23 mL.
7. **Deactivated 4–6 weeks** after implant (tandem: 6 weeks).
8. **Erosion 2–15%** (to 35% with multiple risks); after prior erosion ~50% at 5 years.
9. **Infection or erosion → explant all components**; reimplant 3–6 months, new site.
10. **Failed AUS >3 years old → replace whole device (PRB)**; <3 years → failed part.

CLASSIFICATION, PATHOPHYSIOLOGY, AETIOLOGY

- **Commonest cause: radical prostatectomy (RP)** → external sphincter alone left.
- **Male SUI requires both sphincters impaired** (internal + rhabdosphincter).
- **Prevalence**: 11% at 60–64; 31% at >85 years.
- **Post-RP UI (urinary incontinence)**: 2.5–87%; improves for up to 2 years.
- **Nerve sparing** → faster return, not better final continence.

MECHANISM OF CONTINENCE

- **Transobturator sling repositions**, does not compress → needs functioning sphincter.
- **AUS cuff**: ~2 cm circumferential; void by straining or Credé.
- **Virtue**: prepubic compression + transobturator elevation.
- **ATOMS, ProACT**: saline-adjustable via scrotal port(s).

EVALUATION AND DIAGNOSIS

- **Gold standard severity: 24-h pad weight**; pad counts unreliable.
- **Voiding diary**: 2–7 days.
- **Cough test**: Grade 0 ≈ 57 g → Grade 4 ≈ 513 g; drops → stream.
- **Urinalysis ± culture mandatory**; eradicate urinary tract infection first.
- **Cystoscopy mandatory**: suspected stricture/anastomotic stenosis; after erosion explant.
- **Poorly coapting sphincter → AUS**, not transobturator sling.
- **Urodynamics (UDS) not routine**: mixed, undemonstrable, overflow, low compliance.
- **Hypocontractile detrusor → AUS** over sling.
- **Check manual dexterity** before AUS.
- **Leak during sleep** → severe SUI rather than urgency.
- **SUFU Pad Test**: tracks loss; not validated to predict surgery.

INDICATIONS FOR SURGERY

- **Operate if**: irreversible sphincter incompetence + bothersome UI.
- **Operate without delay**: severe/gravitational SUI, no gain by 6 months.
- **Pelvic floor training**: no benefit beyond 12 months.
- **First-line: AUS or perineal sling**. Bulking agents: not in post-RP algorithm.
- **Absolute (need alternative)**: kidney-threatening low compliance; low capacity; unusable tissue.
- **Unusable tissue** → clamp, catheter, bladder neck closure, diversion.
- **Relative**: bladder cancer, refractory stenosis, radiation cystitis.

- **Radiation cystitis**: hyperbaric oxygen + 6 months bleeding-free before AUS.
- **No AUS**: poor dexterity/cognition, poor urethra, severely low compliance.
- **AUS**: social continence 55–77%; satisfaction ~90%; revision 10–29% at 3 years.
- **AUS life**: 7–10 years (text) vs 8–10 years (guideline).
- **Sling cure ~60%**; ProACT dry ~60% (more erosion); ATOMS dry ~67%.
- **Sling failure predictors**: RT, severity, previous surgery.
- **Counsel**: one thin pad/day; failed sling → AUS.
- **AUS success predictors**: surgeon experience, volume, intact compliance.
- **Advance XP durable**; I-STP TOMS decays over 5 years.
- **Sling in severe SUI** only if AUS refused or unable to use pump.

HIGH-RISK PATIENT

- **Fragile urethra**: RT, prior failed AUS, urethroplasty, hypogonadism.
- **Also**: coronary artery disease; age >80 years.
- **Erosion after RT**: 3.4–33% vs 1–13%; erode sooner.
- **Urethroplasty > RT** as erosion risk.
- **Diabetes, smoking**: not proven independent — optimise anyway.
- **Transcorporal (TC) cuff — Webster, 2002**: tunica interposed dorsally; benefit unproven.
- **Mitigation**: 51–60 PRB; cuff 0.5 cm over measured.
- **RT predicts first explant**, not later revisions.
- **Low testosterone = risk**; replacement not proven protective.

TECHNIQUE OF IMPLANTATION

- **Antibiotics ≤60 min**: aminoglycoside 5 mg/kg + cephalosporin or vancomycin.
- **Skin**: chlorhexidine–alcohol.
- **Cuff 2 cm wide**; sizes 3.5–11 cm; bladder neck ≥8 cm.
- **Cuff tighter than tape**; looser cuffs → better continence.
- **3.5-cm cuff after RT** → more erosion → TC and/or 51–60 PRB.
- **No ring/ectopic PRB**: mesh hernia repair, cystectomy, major abdominal surgery.
- **Contrast diluted with sterile water**; contrast allergy → saline.
- **Closed cuff = slight urethral blanching** on urethroscopy.
- **Tandem**: ≥1.5–2 cm distal, Y-connector, +3 mL; not first-line.
- **TC cuff**: urethra <3.5 cm; more transient retention.
- **Transscrotal — Wilson, 2003**: single incision; inferior continence (smaller cuff).
- **Bladder neck AUS: absolute contraindication after RP**; for intact prostate (e.g. myelomeningocele).
- **"Gullwing"**: complete tunical wrap; no long-term data.

- **Cuff around bulbospongiosus:** erosion rare, revision high.
- **DuraPrep:** fewer skin cultures, same infection rate.

MALE SLINGS

- **Rehder-Gozzi, 2007:** first transobturator male sling.
- **Indications:** SUI after prostate cancer treatment, TURP, laser enucleation.
- **Stab ~1 fingerbreadth below adductor longus;** outside-in; two "pops".
- **Tension to cystoscopic coaptation;** bulb rises 3–4 cm.
- **ATOMS:** bulbospongiosus not opened; port 9 mL.

PERSISTENT INCONTINENCE AFTER AUS OR SLING

- **Sudden → deactivation or leak;** gradual → atrophy.
- **Reactivate:** hold button ≥30 s.
- **PRB >20 mL on imaging →** cuff problem; little/no fluid → leak.
- **Erosion:** urgent explant + catheter ~3 weeks.
- **Failed sling → AUS (gold standard),** results as first-time.
- **Advance loosened ≤6 weeks →** re-explore, replace.
- **Worsening SUI after sling → cystoscopy** (mesh erosion).
- **Second Advance sling:** ~half succeed; only if AUS barred/refused.
- **PRB refills passively, changes size →** atrophy, not leak.
- **Exclude PRB herniation:** needs deep pelvic/submuscular site.

AUS COMPLICATIONS

- **Retention:** catheter 24–48 h → failed trial at 48 h → image-guided suprapubic tube.
- **Retention lasting weeks →** undersized cuff → replace.
- **Infection 1–3% (≤10% high risk);** *S. epidermidis*, *S. aureus*; late >4 months.
- **Salvage contraindicated:** sepsis, ketoacidosis, necrotising infection, erosion, immunosuppression, pus.
- **Post-erosion stricture 8–35%;** <50% urethral loss → catheter.
- **Re-erosion after reimplant:** up to 35%.
- **Atrophy →** downsize or relocate cuff; 4.0-cm cuff → move proximal, or distal TC.
- **Early infection sign:** scrotal pain.
- **Erosion >50% urethral loss:** best repair unresolved.
- **Catheterising an AUS patient =** erosion risk; delayed activation lowers erosion.
- **Reimplant after erosion:** nightly deactivation.
- **Early failures (<3 years):** pump or cuff, median ≤1 year.

SPECIAL CIRCUMSTANCES

- **ROBUST III:** paclitaxel balloon 83.2% vs 21.7% (1-year); 67% at 3 years.

- **22–24 Fr scope past cuff →** uncouple cuff first.
- **Stenosis <12 Fr dilated with AUS →** no excess erosion.
- **Neobladder SUI 33–95%;** PRB ectopic behind rectus.
- **AUS + inflatable penile prosthesis (IPP):** reoperation ~25% either timing; staged → AUS first.
- **Climacturia 20–68%;** mini-jupette = 3-cm corporotomies + graft.
- **Paclitaxel balloon:** bulbar data only; none for stenosis.
- **Refractory anastomotic stenosis →** robotic reconstruction (spares cuff).
- **Stricture before AUS →** non-transecting urethroplasty.
- **Neobladder AUS:** most improve; most need revision.

SLING COMPLICATIONS

- **Non-adjustable:** commonest pain/wound; erosion/explant 0–0.4%; chronic pain 1.3%.
- **Inguinal pain:** self-limiting; median VAS 0 at 3 months.
- **ATOMS:** erosion + infection 10%; explant 10–15.8%.
- **ProACT:** commonest perforation 5.3%; revision 22%.

SUMMARY

- **~90% improved urinary QoL** after surgery; >150,000 men implanted.

CLASSIC TRAPS

- **Trap:** transobturator sling compresses — **Actually:** repositions; needs working sphincter.
- **Trap:** mild UI after RT → sling — **Actually:** any RT → AUS first-line.
- **Trap:** raise PRB pressure for atrophy — **Actually:** erosion; downsize or relocate.
- **Trap:** bladder neck AUS after RP — **Actually:** contraindicated.
- **Trap:** InhibiZone cuts AUS infection — **Actually:** shown for penile implants only.
- **Trap:** keep clean PRB at erosion explant — **Actually:** remove all components.
- **Trap:** salvage infected eroded AUS — **Actually:** erosion bars salvage.
- **Trap:** adjustable devices safer — **Actually:** ATOMS, ProACT erode/revise more.
- **Trap:** large scope through active cuff — **Actually:** uncouple cuff first.
- **Trap:** hypocontractile detrusor → sling — **Actually:** AUS (opens to void).